
The Interactions between Renewable Portfolio Standards and Carbon Emission Trading in China: an Evolutionary Game Theory Perspective

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Abstract: Renewable Portfolio Standards (RPS) and Carbon Emission Trading (CET) are the policy tools for China to achieve the low-carbon energy transition. The rationality of its policy parameter design will affect the selection strategy of market participants, and there is an interactive relationship between markets. Changes in policy parameters will affect the equilibrium state of the interactive system, and bring uncertain risks to the expected effects of policy implementation. Based on the assumption that RPS assessment is undertaken by the power users and CET market is extended to industries other than the power generation industry, this paper constructs a game model under RPS and CET where groups of power users and thermal power manufacturers jointly participate. Then, based on China's provincial market data, it analyses the evolution process of group strategy under different combinations of policy parameters, and described the characteristics of the transition between the stable and unstable states of the system. Finally, a method of dynamic mechanism design and differentiated mechanism design for policy parameters is proposed to restrain system oscillations. The research in this paper provides an analysis framework for China to implement RPS and CET, and provides relevant analysis basis for the setting of policy parameters.

Keywords: Carbon emission trading; Renewable portfolio standards; Evolutionary game; Stability analysis; Policy parameter combination

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1 Introduction

With the increasingly severe global climate change, President Xi has promised China's carbon peaking and carbon neutrality tasks. The development of renewable energy is an important strategic measure to address climate change and achieve carbon peaking and carbon neutrality. Although the cost of renewable energy power generation in China has declined significantly, the technological progress rate based on historical experience is still not enough to support the goal of grid parity for renewable energy. In the future, it is still necessary to further promote the progress of renewable energy technology, the reduction of power generation costs and the improvement of China's renewable energy policy, to promote the sustainable and stable development of renewable energy (TU et al., 2020)^[1]. With the deepening of national electric power system reformation, the renewable energy has gradually shifted from the planned mode to market bidding mode (NDRC, 2021)^[2]. When different types of power supply conduct bidding on the same platform, the current high cost of renewable energy, including capital construction cost, operation and maintenance cost, auxiliary service cost and other costs, makes renewable energy do not have the market bidding advantage in cost (BAO et al., 2019)^[3].

In this context, the Chinese government has issued a series of policies to promote renewable energy consumption, including RPS (NDRC, 2019)^[4] and CET (MEE, 2021)^[5]. In the RPS market, the power users will be the main body of the RPS obligation to complete the renewable energy consumption responsibility weight index, and accept the annual assessment with the actual consumption of renewable energy power as the main method. However, China has not yet issued detailed rules on how power users should undertake the assessment. And in the national CET market, China is currently conducting a pilot carbon emission trading in the power generation industry, and will gradually expand to other high-emission industries.

Under the influence of the above policies, for power generators, renewable energy has unique green attributes,

coupled with the cost of carbon allowances that thermal power companies need to bear, thus increasing the competitiveness of renewable energy in the grid parity model. For power users, under the dual role of RPS and CET, the cost structure has changed from only bearing electricity cost to a combination model of "electricity cost, renewable energy quota and carbon emission quota".

The decision-making behavior of each stakeholder will be affected by policy parameters such as market prices (electricity, carbon emission quota, and RPS quota), RPS requirements, and carbon quota allocation methods. The decision-making behavior of stakeholders will affect the development trend of renewable energy. However, there is great uncertainty as to whether the trend can evolve in accordance with the expected target.

Therefore, it is necessary to study the evolution trend of the decision-making behavior of various stakeholders under different policy parameter combinations, and analyze the stability of the trend to avoid the risk of uncertain market development expectations.

At present, the research on the policy implementation of RPS and CET is mainly carried out from the following aspects:

(1) Empirical analysis based on historical data. Based on the historical data analysis of carbon emissions trading pilots, it is demonstrated that the development of the carbon market has positive effects on environmental health benefits (Zhang and Zhang, 2020)^[6], corporate carbon innovation (Zhu et al., 2019)^[7], employment (Yu and Li, 2021)^[8], carbon emission reduction (Zhang et al., 2019)^[9], air pollutant emission reduction (Yan et al., 2020)^[10] and low carbon international competitiveness (Qi et al., 2021)^[11]. However, differences in the distribution of carbon emission allowances will also bring risks such as regional development inequality and efficiency loss (Wang et al., 2019)^[12]. For the RPS market, Barbose et al. (2016)^[13] demonstrates that the implementation of RPS reduces greenhouse gas emissions while increasing employment. In terms of the interaction between the carbon market and the RPS market, Liu and Jin. (2020)^[14] constructed a quantitative analysis method, identifies the interactions between electricity, fossil fuel and carbon market prices. Schusser and Jaraité. (2018)^[15] confirmed carbon prices can positively affect TGC prices in the short run.

(2) The method based on simulation deduction. Some scholars use the method of system dynamics for simulation deduction. Zhao et al (2017)^[16] and Zhang et al. (2017)^[17] analyzed the influence of RPS system parameters on the development of waste incineration power generation and wind power industries. Feng et al. (2018)^[18] analyzed the impact on China's energy structure under the interactive influence of CET market and RPS market. Zhou et al. (2021)^[19] analyzed the impact of RPS on CET market. In addition, the CGE model is used for analysis from the macroeconomic level. Weng and Xu. (2018)^[20] analyzed the impact of CET on Shanghai's GDP and social welfare. Dai et al. (2018)^[21] analyzed the impact of CET and renewable energy policy on the development of various sectors of the Chinese economy. Chen et al. (2021)^[22] used numerical simulation to analyze the impact of different carbon quota allocation methods on renewable energy and traditional energy investment.

(3) Game-based equilibrium analysis method: In terms of finite individual game, Currier (2013)^[23] constructed the cournot equilibrium model of power oligarchs under the green certificate trading system, and analyzes the optimal percentage requirement. Tanaka and Chen (2013)^[24] constructed an equilibrium analysis model, showed that the behavior of market power in the RPS market will lead to market failure. Sun (2016)^[25] analyzed how to set the required value of RPS of neighboring countries under the TGC market to maximize the social welfare of the two countries. Helgesen and Tomasgard (2018)^[26] analyzed the green certificate and electricity market equilibrium under different market competition modes. In terms of group evolutionary games. Zhao et al. (2018)^[27] analyzed the impact of the relevant institutional quasi-parameters of the quota system on the strategic behavior of power plants. Fang et al. (2019)^[28] proposed an evolutionary game between power companies and regulators under the

RPS system, and proposed a dynamic subsidy penalty mechanism.

(4) Optimization modeling and analysis methods: mainly focus on power investment planning and construction, policy parameter design and the impact on the interests of market participants. Bonenti et al. (2013)^[29] analyzed the issues of power investment planning and construction in the carbon market under the mode of perfect competition and monopolistic competition. Yi et al. (2019)^[30] constructed a multi-regional power optimization model and quantitatively evaluated the impact of RPS and carbon emission caps on China's power planning. Xu et al. (2021)^[31] analyzed the fairness of RPS quota allocation in China's provinces. Pineda and Bock (2016)^[32] analyzed how to set the RPS requirement value as well as the penalty parameter to incentivize the capacity expansion of renewable power generation. Suo et al. (2017)^[33] analyzed the optimal energy supply scheme under CET and green certificate (GC) modes. Guo et al. (2020)^[34] constructed a bi-level optimization model to analyze the strategic behavior of renewable energy under the combined action of the electricity market and the TGC market.

We summarize the above researchers as shown in table 1-1.

Tab 1-1 the summary of researchers

Policy background	Method	Research	Features
Focus on RPS	Empirical analysis based on historical data	Barbose et al., 2016	based on historical data; focus on RPS
	simulation deduction	Zhao et al., 2017; Zhang et al., 2017; Chen et al., 2021	focus on RPS
	Game-based equilibrium analysis	Currier, 2013; Tanaka and Chen, 2013; Sun, 2016; Helgesen and Tomasgard, 2018; Zhao et al., 2018; Fang et al., 2019	focus on RPS
	Optimization modeling	Xu et al., 2021; Pineda and Bock, 2016; Guo et al., 2020	focus on RPS
Focus on CET	Empirical analysis based on historical data	Zhang and Zhang, 2020; Zhu et al., 2019; Yu and Li, 2021; Zhang et al., 2019; Yan et al., 2020; Qi et al., 2021; Wang et al., 2019	based on historical data; Focus on CET
	simulation deduction	Weng and Xu., 2018	focus on CET
	Optimization modeling	Bonenti et al., 2013	focus on CET
Focus on RPS and CET	Empirical analysis based on historical data	Liu and Jin, 2020; Schusser and Jaraitė, 2018	based on historical data
	simulation deduction	1Feng et al., 2018; Zhou et al., 2021	assess the power generation manufacturers
		Dai et al., 2018	macroeconomic analysis
	Optimization modeling	Yi et al., 2019; Suo et al., 2017	individual decision

As shown in table 1-1, the existing researchers mostly focus on the impact of the independent RPS and CET markets, as well as the research on the coupling effect of RPS and CET. In addition, the existing research regards power generation companies as the main body of the RPS and CET market assessment, while in China's policy documents, it is required that the RPS assessment needs to be transmitted to power users, and the CET market needs to be expanded to industries other than the power generation industry.

However, China's RPS and CET markets have not been developed for a long time and there is not yet sufficient data accumulation, the empirical analysis method based on historical data has certain limitations in order to analyze

the impact of policy parameter changes on future market participants' strategic choices. Moreover, the market policy mechanism is constantly updated and iterative. The market participants involved in RPS and CET are massive, and the strategic choices among market participants may not be completely rational, and there is mutual imitation behavior. Therefore, evolutionary games are suitable to solve this problem.

This paper considers the key market participants in the RPS and CET markets, constructs a tripartite evolutionary game for three game agents: power users (participate in electricity market), power users (participate in CET market) and thermal power plant, and analyzes the strategic stability of each game party and the impact of each policy parameter on the strategy choice; secondly, the evolutionary stability strategy under different policy parameters is analyzed. Finally, based on some empirical data in China, the validity of the model analysis under different initial conditions is verified, and based on the analysis conclusions, it provides countermeasures and suggestions for the government to improve the RPS and CET market.

2 Current Situation and Mechanism Analysis

2.1 Current Situation Analysis

In the published rules (NDRC, 2019; MEE, 2021; Weng and Xu, 2018)^{[4][5][35]}, the key points in the RPS and CET are summarized in Table 1-1.

Table 1-1 the key rules in RPS and CET market

	RPS	CET
Quota types	renewable energy quota	carbon emission quota
Quota provision method	Free quota (decrease year by year)	Free quota (benchmark method, historical strength method)
Involved market participates	The rules require: power grid companies, electricity sales companies, and power users participating in the electricity wholesale market. Current: power grid companies	Rules require: power, steel, cement and other key high-emission industries Current: The power generation industry
Supplementary quota method	Consumption above quota, tradable green certificates	carbon emission quota, CCER
Penalty method	Included in bad credit records and joint punishment	Fines and reducing the free quota for the next year

As shown in Table 1-1. The current implementation method of the RPS market is mainly based on the actual renewable energy consumption of each assessment subject. For the unfinished part, the provincial power grid is used as the object to conduct inter-provincial transactions with the consumption above quota. In 2020, the inter-provincial transaction of consumption above quota was officially launched, and 2.455 million certificates were successfully transferred. Through the transfer transaction, all provinces can achieve the national policy requirements (NEA, 2020)^[36] But for market participates, there are no relevant documents for the quota assessment rules. Meanwhile, according to the latest development of the national carbon market, the national CET market takes the power generation industry first, and gradually expands the scope of industries covered, the number of companies and carbon emissions.

With the goal of carbon peaking and carbon neutrality proposed, ETC (2019)^[37] calculated the balance of electricity supply and demand in China in 2050 under the zero-carbon scenario, and pointed out that wind power and photovoltaics will become the main power source, and its power generation will account for 70%. In order to support the demand for a high proportion of renewable energy, most of the wind turbines in the future will be built in areas with abundant wind and wind resources such as Inner Mongolia, Xinjiang, and provinces in the Northeast Region (Li et al., 2021).^[38] Under such a rigid carbon neutrality target constraint, the geographical distribution of different non-hydro renewable energy sources will bring about the liquidity of renewable energy quota market transactions, and through the spontaneous economic behavior of market participates, the allocation efficiency of

power generation resources will be improved.

Therefore, the scenario of this paper assumes that the main body of the RPS assessment will be transferred to the end power users in the future, and the national CET market will gradually expand to industries other than the power generation industry. A tripartite evolutionary game model is constructed to analyze the market equilibrium state under different policy parameter combinations. .

2.2 Mechanism Analysis

(1) The mechanism of the RPS market

In the context of China's policy, based on energy planning and the resource endowment of each province, the relevant departments will roll out the proportion of non-hydro RPS at the beginning of each year, showing a trend of increasing year by year. Through the constraints of RPS, it is encouraged to improve the consumption level of local renewable energy, promote the trans-provincial transaction of renewable energy, and realize the optimal allocation of renewable energy in a wider range. From the perspective of user electricity cost, market participants need to undertake a certain proportion of renewable energy consumption obligations when making electricity decisions, otherwise they will face certain assessments.

(2) The mechanism of the CET market

In the initial stage of the national CET market, the allocation plan for carbon emission quota in the power generation industry is the industry benchmark method, which implements part of free allocation. If the industry's baseline value is low, companies that exceed the baseline will bear greater pressure to reduce emissions. Under the baseline trading mechanism, the total amount of free carbon emission quota obtained by power generating units with different carbon emission intensities will vary with their own power generation. Without considering the marginal cost of generating units, when the actual carbon emissions per unit power generation of the generating units are greater than the quota baseline approved by the regulatory agency, the high-emitting unit needs to buy carbon emission quota corresponding to its own power generation in the CET market. On the contrary, the low-emission unit can sell the surplus carbon quota corresponding to its own electricity in the CET market to obtain a part of the income.

(3) Mechanism of interaction between RPS and CET markets

Under the combined effect of the above two markets, the relevant market participants include power users participating in the wholesale electricity market, thermal power manufacturers, and non-hydro renewable energy manufacturers (hereinafter referred to as "green power manufacturers"). In order to fully analyze the effect of the CET market, the analysis object in this paper is the market entities whose carbon emission reduction efficiency is lower than the industry benchmark value. These market players need to purchase carbon quota or undertake assessments. The interest composition and decision-making behavior of power users, thermal power manufacturers and green power manufacturers in the above-mentioned markets are analyzed as follows:

For thermal power manufacturers, they need to consume carbon quotas when producing electricity. According to the analysis of the market supply and demand relationship, the electricity price P_e decreases with the increase of electricity production, and the carbon market price P_c increases with the increase of carbon quota demand. The average cost of thermal power plants P_{ac} gradually decreases with the increase of output. As shown in Figure 2-1, in the market competition environment, when the electricity production revenue R is greater than the carbon allowance price, thermal power manufacturers will choose to produce; otherwise, they will choose not to produce electricity, or bear the corresponding risk of allowance assessment.

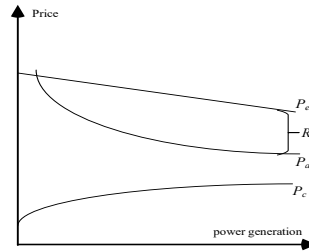


Fig 2-1 the revenue and cost composition of thermal power manufacturers

For green power manufacturers, since the variable cost of green power is extremely low, the main cost consists of initial investment, operation and maintenance management fees. The main factor affecting the cost of electricity is utilization hour, and there is no carbon quota constraint. Therefore, if power users choose to trade with green power manufacturers, the green power manufacturers choose to trade directly. At this time, green power manufacturers receive electricity revenue and the TGC revenue.

For power users, it undertakes the quota assessment requirements in RPS and CET markets. In the RPS market, a certain proportion of green electricity needs to be included in the electricity consumption of users, which can be accomplished by consuming non-hydro renewable energy, purchasing excess consumption from other market participants, and purchasing TGC. In the CET market, if the amount of carbon quota used by power users exceeds the free quota, a carbon market trading decision is required: if the decision requires the purchase of carbon quota, it will form a competitive relationship with thermal power plants for carbon quota in CET market; if the decision does not require the purchase of carbon quota, it will need to accept the assessment penalty.

In summary, under the dual role of the RPS and CET market, the interaction between various market participants is shown in figure 2-2:

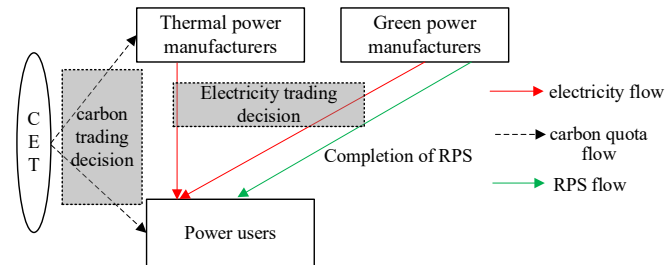


Fig 2-2 Interaction Framework for RPS and CET Markets

As shown in Figure 2-2, there are two key decision points among the three types of market participants. The first is decision-making in electricity market, the final transaction electricity component of power users will affect their RPS performance. The second is carbon trading decision-making. Changes in the share of thermal power generation will affect the price of the CET market. For example, if the share of thermal power output is large, thermal power manufacturers and power users will jointly form a demand for carbon quota, which will affect the CET market price. In addition, there are also market participants who choose not to perform the contract and choose to bear the corresponding punishment.

3 Modeling

3.1 Assumptions

This paper analyzes the impact of different policy parameters on the decision-making behavior of stakeholders when the RPS assessment is finally transmitted to the user side and the carbon market expands to industries other than the power generation industry.

The original problem is a two-stage decision problem. First, power users make the decision of electricity trading (choose to purchase thermal power or green power), and then enter the second stage of the game between power users and thermal power manufacturers (choose whether to participate in the CET market). Since the first stage only involves the selection of power users, its selection strategy needs to be formulated according to the results of the game in the second stage, and the power users in the first and second stages have the same profit matrix. There are three types of market participants. The electricity users who participate in the decision-making of the electricity market in the first stage are regarded as the first group, named as "power users (electricity market)", and the electricity users who participate in the CET market in the second stage are regarded as the second group, named as "power users (CET market)", and the thermal power manufacturers in the second stage are regarded as the third group, named as "thermal power manufacturers".

Therefore, this two-stage decision problem can be transformed into a static tripartite game problem, and its logical relationship is shown in Figure 3-1.

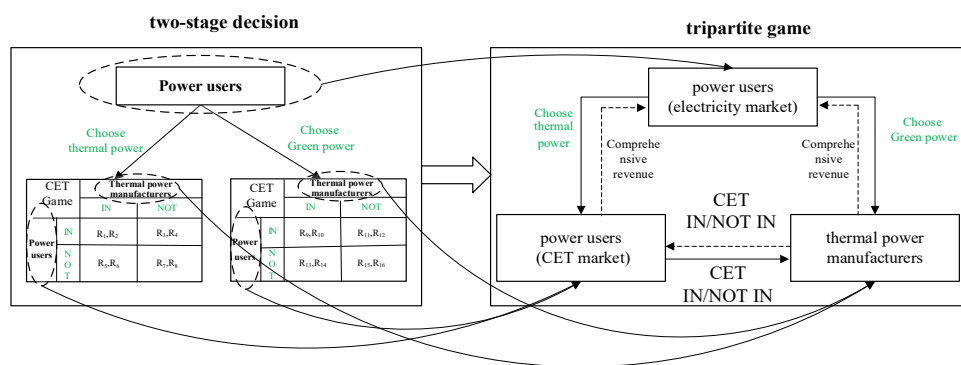


Fig 3-1 Game framework

In order to focus on analyzing the evolution of the behavior of market participants under the joint action of the RPS and CET markets, the following assumptions are made:

Assumption 1: There are three types of groups involved in the model. Group 1: power users (electricity market), the proportion of the group that chooses to buy green electricity is x , and the proportion of the group that chooses to buy thermal power is $1-x$. Group 2: power users (CET market), the proportion of the group that chooses to participate in the CET market is y , and the proportion of the group that chooses not to participate in the CET market is $1-y$. Group 3: Thermal power manufacturers, the proportion of the group that chooses to participate in the CET market is z , and the proportion of the group that chooses not to participate in the CET market is $1-z$.

Assumption 2: The RPS assessment is conducted to the power consumption side, that is, the power user in the model undertakes RPS the assessment, and the consumption of the power user requires a certain proportion of renewable energy (NDRC, 2019) [4]. The assessment subjects of the CET market include both thermal power manufacturers and power users (MEE, 2021)[5].

Assumption 3: The price of carbon quota is affected by market demand. Both thermal power manufacturers and power users have a demand for the carbon quota, and they are in a competitive relationship with each other. To simplify the analysis, the carbon market demand will be divided into three levels: high, medium and low. If one of the power users and thermal power manufacturers chooses to participate in the carbon market, the demand at this time is medium demand, and the corresponding carbon price is P_i . If both power users and thermal power manufacturers choose to participate in the carbon market, when power users choose to buy green power, the power generation of thermal power manufacturers is lower than when power users choose to buy thermal power, so the

241 demand for carbon quota is also different. Therefore, it can be further divided: 1) When power users purchase green
 242 electricity, corresponding to the demand for medium carbon quota, and the corresponding carbon price is P_m ; 2)
 243 When power users purchase thermal electricity, corresponding to high carbon quota, and the corresponding carbon
 244 price is P_h , with $P_l < P_m < P_h$.

245 **Assumption 4:** The carbon emission reduction efficiency of both power users and thermal power
 246 manufacturers is lower than the industry benchmark value, that is, thermal power needs to purchase carbon quota
 247 for power generation, and its purchase ratio is β , and power users need to purchase carbon quota for business
 248 production with a proportion of θ .

249 **Assumption 5:** If market participants fail to meet the RPS or carbon emission requirements, and fail to make
 250 up for it in time, they need to be punished. The fine for the RPS per unit of electricity is F_{RPS} , and the fine for each
 251 unit of carbon quota is F_c .

252 The specific parameters are shown in Table 3-1.

253

Tab 3-1 Model parameters		
symbol	definition	unit
I_e	unit electricity output value of power users	Yuan/MWh
Q_e	power demand of electricity users	MWh
E_t	carbon emissions of thermal power manufacturers	ton/MWh
E_e	carbon emissions per unit of output value of power users	ton/yuan
α	RPS ratio requirement	%
β	The proportion of carbon quota that thermal power manufacturers need to purchase	%
θ	The proportion of carbon quota that power users need to purchase	%
P_t	On-grid electricity price of thermal power	yuan/MWh
C_t	Thermal power unit cost	yuan/MWh
P_g	On-grid electricity price of green power	yuan/MWh
$P_{l,m,h}$	Low, medium and high levels of carbon price	yuan/ton
F_{RPS}	Fine for RPS market	yuan/MWh
F_c	Fine for CET market	yuan/ton

254 **3.2 Model payout matrix**

255 According to the assumption of the model, the payment function of power users consists of operating income
 256 of power users, expenditures for purchasing electricity, expenditures for purchasing carbon quota, and assessment
 257 expenditures for RPS and CET. The payment function of thermal power manufacturers consists of electricity
 258 production revenue, power generation cost expenditure, carbon quota purchase expenditure and CET assessment
 259 expenditure. Since power users (electricity market) and power users (CET market) are the same market participant,
 260 the payment function is the same. The payment function is shown in Table 3-2.

261

Tab 3-2 game payoff matrix				
			thermal power manufacturers	
			in z	not in $1-z$
power users (electricity market)	green power	in	$U_1 = Q_e I_e - Q_e P_g - \theta Q_e I_e E_e P_m$	$U_2 = Q_e I_e - Q_e P_g - \theta Q_e I_e E_e P_l$
		y	$U_1 = Q_e I_e - Q_e P_g - \theta Q_e I_e E_e P_m$	$U_2 = Q_e I_e - Q_e P_g - \theta Q_e I_e E_e P_l$
	x	not in	$U_3 = Q_e I_e - Q_e P_g - \theta Q_e I_e E_e F_c$	$U_4 = Q_e I_e - Q_e P_g - \theta Q_e I_e E_e F_c$
		1-y	$U_9 = -\beta Q_e E_t P_m$	$U_{10} = -\beta Q_e E_t P_l$

thermal power 1-x	in	$U_5 = Q_e I_e - Q_e P_i - \theta Q_e I_e E_e P_h - \alpha Q_e F_{RPS}$	$U_6 = Q_e I_e - Q_e P_i - \theta Q_e I_e E_e P_i - \alpha Q_e F_{RPS}$
		$U_5 = Q_e I_e - Q_e P_i - \theta Q_e I_e E_e P_i - \alpha Q_e F_{RPS}$	$U_6 = Q_e I_e - Q_e P_i - \theta Q_e I_e E_e P_i - \alpha Q_e F_{RPS}$
	y	$U_{11} = Q_e P_i - \beta Q_e E_e P_h - Q_e C_i$	$U_{12} = Q_e P_i - \beta Q_e E_e F_c - Q_e C_i$
	not in	$U_7 = Q_e I_e - Q_e P_i - \theta Q_e I_e E_e F_c - \alpha Q_e F_{RPS}$	$U_8 = Q_e I_e - Q_e P_i - \theta Q_e I_e E_e F_c - \alpha Q_e F_{RPS}$
	1-y	$U_{13} = Q_e P_i - \beta Q_e E_e P_i - Q_e C_i$	$U_{14} = Q_e P_i - \beta Q_e E_e F_c - Q_e C_i$

4 Model analysis

4.1 Stability Analysis of Power User's Electricity Selection Strategy

The total expected benefit of electricity users when they choose to purchase green electricity is:

$$Ex = yzU_1 + y(1-z)U_2 + (1-y)zU_3 + (1-y)(1-z)U_4$$

The total expected benefit when electricity users choose to purchase thermal electricity is::

$$E(1-x) = yzU_5 + y(1-z)U_6 + (1-y)zU_7 + (1-y)(1-z)U_8$$

From the above, it can be obtained that the comprehensive expected benefit of the power user's electricity selection strategy is $\bar{Ex} = xEx + (1-x)E(1-x)$.

In order to analyze the dynamic change rule of the population proportion using a certain strategy, it is necessary to obtain the replication dynamic equation of the strategy, that is $dx_1(t)/dt = (u_1 - \bar{u})x_1(t)$. The replication dynamics depends on the population proportion of the strategy on the one hand, and on the difference between the strategy's benefit and the population's average benefit on the other hand (Xie, 2019) [39]. Then the replica dynamic equation of the power user's electricity selection strategy is expressed as follows:

$$F(x) = x(Ex - \bar{Ex}) = x(1-x)(Ex - E(1-x)) \\ = x(1-x)Q_e [P_i - P_g + \alpha F_{RPS} + yz\theta E_e I_e (P_h - P_m)] \quad (1)$$

Taking the partial derivative of x in the replication dynamic equation gives:

$$\frac{d(F(x))}{dx} = (1-2x)Q_e [P_i - P_g + \alpha F_{RPS} + yz\theta E_e I_e (P_h - P_m)]$$

According to the stability theorem of differential equations, the proportion of power users who choose to purchase green electricity in a stable state must satisfy $F(x)=0$ and $d(F(x))/dx < 0$.

When $x=0$ or 1 , $F(x)=0$, let $G(y) = P_i - P_g + \alpha F_{RPS} + yz\theta E_e I_e (P_h - P_m)$, the sign of $d(F(x))/dx$ depends on the size of $G(y)$.

Taking the partial derivative of y in $G(y)$, we get $d(G(y))/dy = z\theta E_e I_e (P_h - P_m)$, so $G(y)$ is an increasing function of

y . Let $G(y)=0$, we get $y^* = \frac{P_g - P_i - \alpha F_{RPS}}{z\theta E_e I_e (P_h - P_m)}$, and there are:

(1) when $y=y^*$, x is in a stable evolution state.

(2) when $y > y^*$, then $G(y) > 0$, $\frac{d(F(x))}{dx}|_{x=1} < 0$, $x=1$ is an evolutionary stable strategy.

(3) when $y < y^*$, then $G(y) < 0$, $\frac{d(F(x))}{dx}|_{x=0} < 0$, $x=0$ is an evolutionary stable strategy.

According to the above analysis, it can be concluded that when y increases or z increases, the electricity selection strategy of power users evolves from $x=0$ to $x=1$, indicating that when the proportion of power users and thermal power manufacturers participating in the CET market increases, the proportion of power users who choose to purchase green electricity will also increase. That is to say, the enthusiasm of market participants to participate in CET market will increase, and the proportion of market players who choose to purchase green electricity will also increase. The evolution phase diagram of the power user's electricity selection strategy is shown in Figure 4-1.

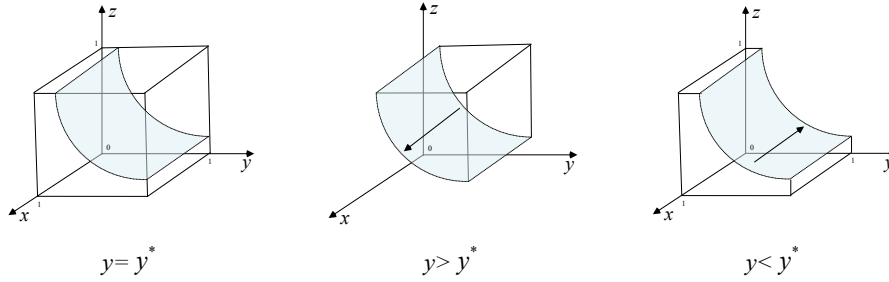


Fig 4-1 Evolution phase diagram of power user's electricity selection strategy

4.2 Stability Analysis of Power Users' CET Selection Strategy

Similarly, the replication dynamic equation for power users to choose to participate in the CET market strategy is:

$$F(y) = y(Ey - \bar{E}y) = y(1-y)(Ey - E(1-y)) = y(1-y)\theta E_e Q_e I_e [F_c - P_l - z(P_h - P_l) + xz(P_h - P_m)] \quad (2)$$

Taking the partial derivative for y in the replica dynamic equation gives:

$$\frac{d(F(y))}{dy} = (1-2y)\theta E_e Q_e I_e [F_c - P_l - z(P_h - P_l) + xz(P_h - P_m)]$$

According to the stability theorem of differential equations, the proportion of power users who choose to participate in CET market is in a stable state must satisfy $F(y)=0$ and $d(F(y))/dy < 0$, when $y=0$ or 1 , then $F(y)=0$, the sign of $d(F(y))/dy$ depends on the size of $G(x) = F_c - P_l - z(P_h - P_l) + xz(P_h - P_m)$. Taking the partial derivative of x in $G(x)$, we get $d(G(x))/dx = z(P_h - P_m)$, so $G(x)$ is an increasing function of x . Let $G(x)=0$, we get $x^* = \frac{z(P_h - P_l) - F_c + P_l}{z(P_h - P_m)}$, and there are:

(1) when $x = x^*$, y is in a stable evolution state.

(2) when $x > x^*$, then $G(x) > 0$, $\frac{d(F(y))}{dy}|_{y=1} < 0$, $y=1$ is an evolutionary stable strategy.

(3) when $x < x^*$, then $G(x) < 0$, $\frac{d(F(y))}{dy}|_{y=0} < 0$, $y=0$ is an evolutionary stable strategy.

Since $\frac{dx^*}{dz} = \frac{F_c - P_l}{z^2(P_h - P_m)} > 0$, then x^* is an increasing function of z .

According to the above analysis, when x increases or z decreases, the CET market selection strategy of power users gradually evolves from $y=0$ to $y=1$, indicating that the proportion of power users purchasing green power increases or the proportion of thermal power manufacturers participating in CET market decreases, which will increase the proportion of market players participating in CET market. The evolution phase diagram of the CET market participation strategy of power users is shown in figure 4-2.

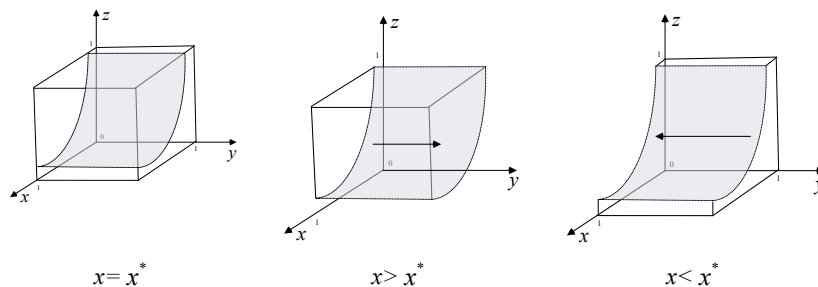


Fig 4-2 Evolution phase diagram of the CET market participation strategy of power users

4.3 Stability Analysis of Thermal Power Manufacturers' CET Selection Strategy

Similarly, the replication dynamic equation for thermal power manufacturers to choose to participate in the CET market strategy is:

$$F(z) = z(Ez - \overline{Ez}) = z(1-z)(Ez - E(1-z)) \\ = z(1-z)\beta E_t Q_e [(1-x)F_c + (y-1)P_l + xy(P_h - P_m) - yP_h] \quad (3)$$

Taking the partial derivative for z in the replica dynamic equation gives:

$$\frac{d(F(z))}{dz} = (1-2z)\beta E_t Q_e [(1-x)F_c + (y-1)P_l + xy(P_h - P_m) - yP_h]$$

According to the stability theorem of differential equations, the proportion of thermal power manufacturers choose to participate in CET market is in a stable state must satisfy $F(z)=0$ and $d(F(z))/dz < 0$, when $z=0$ or 1 , then the sign of $d(F(z))/dz$ depends on the size of $G(y) = (1-x)F_c + (y-1)P_l + xy(P_h - P_m) - yP_h$. Taking the partial derivative of y in $G(y)$, we get $d(G(y))/dy = x(P_h - P_m) - (P_h - P_l)$, since $x < 1$ and $P_h - P_l > P_h - P_m$, so $G(y)$ is a decreasing function of y . Let $G(y)=0$, then $y^* = \frac{P_l - F_c + xF_c}{(P_l - P_h) + x(P_h - P_m)}$, and there are:

(1) when $y = y^*$, z is in a stable evolution state.

(2) when $y > y^*$, then $G(x) < 0$ and $\frac{d(F(y))}{dy}|_{z=0} < 0$, $z=0$ is an evolutionary stable strategy.

(3) when $y < y^*$, then $G(x) > 0$ and $\frac{d(F(y))}{dy}|_{z=1} < 0$, $z=1$ is an evolutionary stable strategy.

Since $\frac{dy}{dx} = \frac{P_l(F_c - P_h + P_m) - F_c P_m}{[P_l - P_h + x(P_h - P_m)]^2}$, $P_l < P_m$, $F_c - (P_h - P_m) < F_c$, then $dy/dx < 0$, so y is a decreasing function of x .

According to the above analysis, When x decreases and y decreases, the proportion of thermal power manufacturers participating in CET market gradually evolves from $z=0$ to $z=1$. On the one hand, due to the reduction in the proportion of power users who choose to purchase green power, the electricity demand of power users is shifted from green power to thermal power, which will increase the carbon emissions demand of thermal power, thus increasing the proportion of thermal power manufacturers participating in CET market. On the other hand, the proportion of power user groups participating in the carbon market decreases, which reduces the demand for CET market and reduces the risk of rising costs for thermal power producers due to rising carbon prices.

The evolution phase diagram of the CET market participation strategy of thermal power manufacturers is shown in figure 4-3.

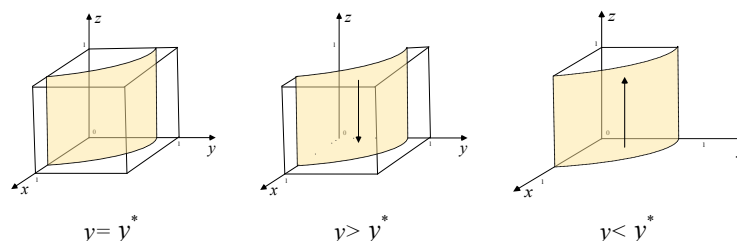


Fig 4-3 Evolution phase diagram of the CET market participation strategy of thermal power manufacturers

4.4 Stability Analysis of System Equilibrium

Let $F(x) = 0, F(y) = 0, F(z) = 0$, we get 12 equilibrium solutions: $E_1(0,0,0)$, $E_2(0,0,1)$, $E_3(1,0,0)$, E_4

$$\begin{aligned}
& (0,1,0), \quad E_5(1,0,1), \quad E_6(0,1,1), \quad E_7(1,1,0), E_8(1,1,1), \quad E_9\left(1, \frac{P_l}{P_l - P_m}, \frac{F_c - P_l}{P_m - P_l}\right), \quad E_{10}\left(0, \frac{F_c - P_l}{P_h - P_l}, \frac{F_c - P_l}{P_h - P_l}\right), \\
& E_{11}\left(\frac{F_c - P_h}{F_c - P_h + P_m}, 1, \frac{P_g - P_l - \alpha F_{RPS}}{\theta E_e I_e (P_h - P_m)}\right), \quad E_{12}\left(\frac{P_h - F_c}{P_h - P_m}, \frac{P_g - P_l - \alpha F_{RPS}}{\theta E_e I_e (P_h - P_m)}, 1\right), \quad E_{13}(e_1, e_2, e_3), \quad E_{14}(e_4, e_5, e_6).
\end{aligned}$$

Since x, y, z all belong to $[0,1]$, and $\frac{P_l}{P_l - P_m} < 0$, so E_9 is meaningless. $E_{10}-E_{14}$ make sense under certain conditions. and its eigenvalue general term is too complicated, the numerical analysis results are shown in Appendix B. The Jacobian matrix of the tripartite evolutionary game is shown in equation (4):

$$\begin{aligned}
J &= \begin{bmatrix} \frac{\partial F(x)}{\partial x} & \frac{\partial F(x)}{\partial y} & \frac{\partial F(x)}{\partial z} \\ \frac{\partial F(y)}{\partial x} & \frac{\partial F(y)}{\partial y} & \frac{\partial F(y)}{\partial z} \\ \frac{\partial F(z)}{\partial x} & \frac{\partial F(z)}{\partial y} & \frac{\partial F(z)}{\partial z} \end{bmatrix} \\
&= \begin{bmatrix} (1-2x)(P_l - P_g + \alpha F_{RPS} + \theta yzP_h - \theta yzP_m) & -\theta xz(P_h - P_m)(x-1) & -\theta xz(P_h - P_m)(x-1) \\ -\theta yz(P_h - P_m)(y-1) & (1-2y)(P_l - P_g + \alpha F_{RPS} + \theta xzP_h - \theta xzP_m) & -\theta xy(P_h - P_m)(y-1) \\ \beta z[F_c - y(P_h - P_m)](z-1) & -\beta z(z-1)[P_l - P_h + x(P_h - P_m)] & \beta(2z-1)(P_l - F_c + xF_c + yP_h - yP_l - xyP_h + xyP_m) \end{bmatrix} \quad (4)
\end{aligned}$$

Based on Lyapunov's indirect method: all eigenvalues of the Jacobian matrix have negative real parts, then the equilibrium point is asymptotically stable; if at least one of the eigenvalues of the Jacobian matrix has a positive real part, then the equilibrium point is not a stable point; except for the eigenvalues with zero real part, the Jacobian matrix has negative real parts, the equilibrium point is in a critical state, and the stability cannot be determined by the sign of the eigenvalues. The eigenvalues of the equilibrium points are shown in table 4-1.

Tab 4-1 Equilibrium point eigenvalues

Equilibrium point	eigenvalues ($\lambda_1, \lambda_2, \lambda_3$)	stability
E_1	$\theta E_e I_e Q_e (F_c - P_l), \beta E_t Q_e (F_c - P_l), Q_e (P_l - P_g + \alpha F_{RPS})$	saddle point
E_2	$\theta E_e I_e Q_e (F_c - P_h), \beta E_t Q_e (P_l - F_c), Q_e (P_l - P_g + \alpha F_{RPS})$	to be determined
E_3	$\theta E_e I_e Q_e (F_c - P_l), -Q_e (P_l - P_g + \alpha F_{RPS}), -\beta E_t Q_e P_l$	saddle point
E_4	$\theta E_e I_e Q_e (P_l - F_c), \beta E_t Q_e (F_c - P_h), Q_e (P_l - P_g + \alpha F_{RPS})$	to be determined
E_5	$\theta E_e I_e Q_e (F_c - P_m), -Q_e (P_l - P_g + \alpha F_{RPS}), \beta E_t Q_e P_l$	saddle point
E_6	$\theta E_e I_e Q_e (P_h - F_c), \beta E_t Q_e (P_h - F_c), Q_e (P_l - P_g + \alpha F_{RPS}) + \theta E_e I_e Q_e (P_h - P_m)$	to be determined
E_7	$\theta E_e I_e Q_e (P_l - F_c), -Q_e (P_l - P_g + \alpha F_{RPS}), -\beta E_t Q_e P_m$	to be determined
E_8	$\theta E_e I_e Q_e (P_m - F_c), \beta E_t Q_e P_m, -Q_e (P_l - P_g + \alpha F_{RPS}) + \theta E_e I_e Q_e (P_h - P_m)$	saddle point

The stability analysis of each equilibrium point is as follows:

(1) Since $F_c - P_l > 0$, then eigenvalues λ_1 in E_1 and E_3 are greater than 0, which are saddle point.

(2) The eigenvalue in E_5 and the eigenvalue in E_8 are greater than 0, which are saddle point.

(3) The eigenvalue general expressions of $E_{10}-E_{14}$ are too complicated, and the analysis will be carried out after the subsequent values are introduced.

(4) The stability of the equilibrium points E_2, E_4, E_6 , and E_7 needs to be further analyzed according to the parameter values. The following judgment conditions exist:

condition ①: $F_c - P_h < 0$;

condition ②: $F_c - P_h > 0$;

condition ③: $P_l - P_g + \alpha F_{RPS} > 0$;

condition ④: $P_l - P_g + \alpha F_{RPS} < 0$;

condition ⑤: $(P_l - P_g + \alpha F_{RPS}) + \theta E_e I_e (P_h - P_m) < 0$.

Among them, ① and ② are mutually exclusive, and ③ is mutually exclusive with ④ and ⑤. The combination of conditions for the equilibrium steady state of the system is shown in Table 4-2.

Tab 4-2 Equilibrium Stability Condition

equilibrium	condition combination
$E_2(0,0,1)$	①④
$E_4(0,1,0)$	①④
$E_6(0,1,1)$	②⑤
$E_7(1,1,0)$	③

The above system stability results and mutual exclusion conditions show that, there is an obvious mutual exclusion relationship between thermal power manufacturers' choice of CET strategy and power users' choice of green power strategy. In CET market, the CET strategies of thermal power manufacturers and power users are also mutually exclusive, and it is necessary to have a stable state in which power users and thermal power manufacturers participate in the CET market at the same time in a more severe environment, which should further satisfy ⑤ on the basis of ④.

The analysis of electricity market selection strategy and CET market selection strategy is as follows:

(1)Electricity market: The green electricity selection strategy of power users is mainly affected by green power price, thermal power price and RPS penalty. When the RPS penalty is greater than the price difference between green power and thermal power, the evolution of the strategy will gradually tend to (buying green power, participating in the carbon market, not participating in the carbon market).

(2) Carbon market: The selection strategy of CET market is mainly affected by factors such as carbon market punishment, carbon price, green power price, thermal power price and RPS requirement value. 1) When the penalty price in- CET market is lower than the highest carbon quota price, and when the RPS penalty is less than the difference between the price of green power and thermal power, the evolution of the strategy will evolve in two directions with the different proportions of the initial population, these two directions are (0,0,1) and (0,1,0). 2) When the penalty in CET market is very large, and the RPS penalty is smaller than the difference between green power and thermal power prices, and the carbon price fluctuation is small, the evolution of the strategy portfolio will gradually tend to (0, 1, 0).

5 Model Simulation

5.1 Parameter settings

Based on the parameter composition of the replication dynamic equation of each group in equations (1) (2) (3). RPS requirements, on-grid price of thermal power, on-grid price of photovoltaic and other parameters are derived from relevant policy documents from 2019 to 2021. Penalties in the carbon market are set at 1-3 times the carbon price. Thermal power companies choose conventional units above 300MW, the carbon emission benchmark value is 0.887t/MWh, the industry average is 304.9g/kWh, the converted carbon emission is 1.118t/MWh. That is, the industry average is still 20.6% behind the benchmark. The carbon market assessment standards for power users refer to the requirements of the Shanghai carbon trading rules, and the proportion of carbon quota directly issued for free is 93% to 99%. For power users, the electrolytic aluminum industry data is selected as the analysis object. The carbon price is derived from the data of the national carbon trading market from July 16, 2021 to August 31, 2021. The parameters are shown in Table 5-1.

Tab 5-1 parameter settings

symbol	parameter value	data source
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α	as shown in appendix C	NDRC and NEA, 2021
β	20.6%	CEC, 2021
θ	5%	SEE, 2020; TONG et al., 2019
P_t	as shown in appendix C	NDRC, 2021
P_g	as shown in appendix C	NDRC, 2019
$P_{l,m,h}$	44, 49.3, 61.07	the national carbon trading market
F_{RPS}	2 times the price of green certificate	Zhang et al., 2017
F_c	2 times the price of carbon quota	Shi et al., 2018

5.2 Simulation analysis

Based on the above parameter settings, since the RPS in China takes provincial administrative units as the assessment object, this paper takes each provincial administrative body in China as the analysis object. The stability conditions for the evolution of strategies in each province are shown in table 5-2.

Tab 5-2 Analysis on the stability of strategy evolution in each province

grid region	province	condition					equilibrium points
		①	②	③	④	⑤	
North China	Beijing		√	√			E_7
	Tianjin		√	√			E_7
	south Hebei		√		√		None
	north Hebei		√	√			E_7
	Shanxi		√	√			E_7
	Shandong		√		√		None
Inner Mongolia	western inner Mongolia		√	√			E_7
	eastern inner Mongolia		√		√		None
Northeast China	Liaoning		√	√			E_7
	Jilin		√	√			E_7
	Heilongjiang		√	√			E_7
	Shanghai		√		√		None
East China	Jiangsu		√		√		None
	Zhejiang		√		√		None
	Anhui		√		√		None
	Fujian		√		√		None
Central China	Jiangxi		√		√		None
	Henan		√		√		None
	Hubei		√		√		None
	Hunan		√	√			E_7
	Chongqing		√		√		None
	Sichuan		√	√			E_7
Northwest China	Shaanxi		√	√			E_7
	Gansu		√	√			E_7
	Qinghai		√	√			E_7
	Ningxia		√	√			E_7
	Xinjiang		√		√		None
south China	Guangdong		√		√		None
	Guangxi		√		√		None

Hainan	√	√	None
Guizhou	√	√	None
Yunnan	√	√	None

It can be seen from Table 5-2 that based on the current parameter settings, there is a stable state E7 (1,1,0), and the provinces with a stable state are mostly located in three Regions (North China, Northeast China and Northwest China), whose non-hydro renewable energy resources are abundant and have great potential for development. According to the analysis of the stability condition ③, the first reason is that the RPS requirement value of the province is relatively high, and the second reason is the large difference between the on-grid electricity price of thermal power and non-hydro renewable energy. As shown in figure 5-1, except for Sichuan Province, which has a very small electricity price difference, most of the other provinces with a stable state have a RPS requirement value.

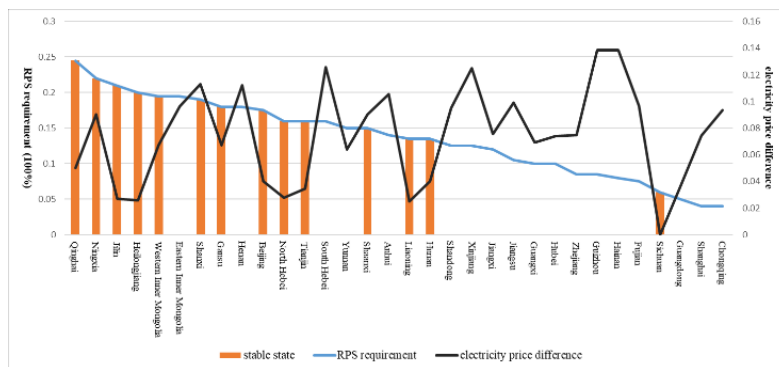


Fig 5-1 RPS requirement value and electricity price difference of each province

From the evolution results (see Appendix A), the green power selection strategies of power users and the carbon trading strategies of thermal power manufacturers in an unstable state both show repeated fluctuations to varying degrees. Frequent changes in the proportion of strategic choices of different groups in market participants will bring uncertainty about the expected effect of policy implementation. Therefore, when government departments formulate relevant policy parameter constraints, they need to set different parameters according to the differences in each province. For example, in order to promote green power consumption, the weight value of non-hydro renewable energy power consumption responsibility can be appropriately increased. With the progress of green power technology and the gradual reduction of production costs, the weight value of consumption responsibility can be reduced to achieve the same value. Effect; In order to enhance the enthusiasm of market participants in some industries to participate in the CET market, then the parameters can be configured according to the stability conditions of the evolutionary equilibrium points $E_2(0,0,1)$, $E_4(0,1,0)$ and $E_6(0,1,1)$.

6 Sensitivity analysis

We select a stable point and an unstable point from table 5-2, and analyze the strategy evolution trend of the market subject group after changing some parameters. According to the parameter settings in table 3-1, the parameters formulated by the government departments include the RPS requirement value, the RPS penalty value, free carbon quota for power users, and the CET penalty value.

6.1 Sensitivity analysis of unstable point

For unstable points, we use the basic parameters of South Hebei in table 5-2 for sensitivity analysis.

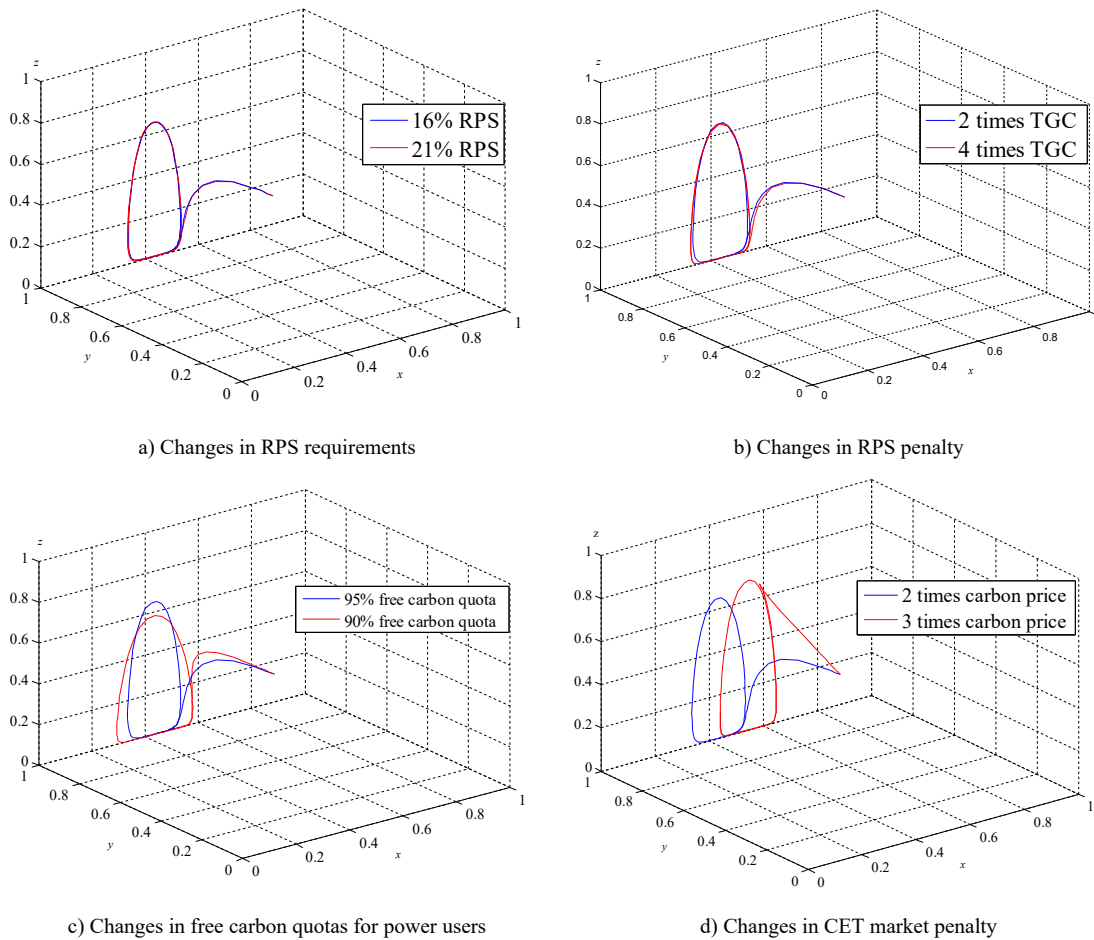


Fig 6-1 Sensitivity analysis of unstable points

It can be seen from Figure 6-1 a) and b) that changing the RPS-related parameters has little effect on the evolution state of the system. It can be seen from Figure 6-1 c) that reducing the proportion of free carbon quotas for power users will increase the fluctuation of power users' green electricity purchase strategy, which is reflected in the larger x-axis amplitude, at the same time, the variation range of thermal power carbon trading strategies is reduced, which is reflected in the smaller z-axis amplitude. It can be seen from Figure 6-1 d) that increasing the penalty in CET market can increase the proportion of electricity users' green electricity purchase strategies, which is reflected in the right shift of the x-axis, at the same time, the change range of the thermal power carbon trading strategy is increased, which is reflected in the larger z-axis amplitude.

The above analysis shows that if the system is in an unstable state, changing the policy parameters of the RPS has no significant guiding effect on the strategic choice of market participants. Changing the policy parameters of the CET market can have a certain impact on the strategic choices of market participants, but the overall instability of the system still exists.

6.2 Sensitivity analysis of stable point E_7

For the basic parameters of Jilin Province in Table 5-2, the system has an evolutionary equilibrium state E_7 (1,1,0) under the current state parameters, the sensitivity analysis results are shown in Figure 6-2.

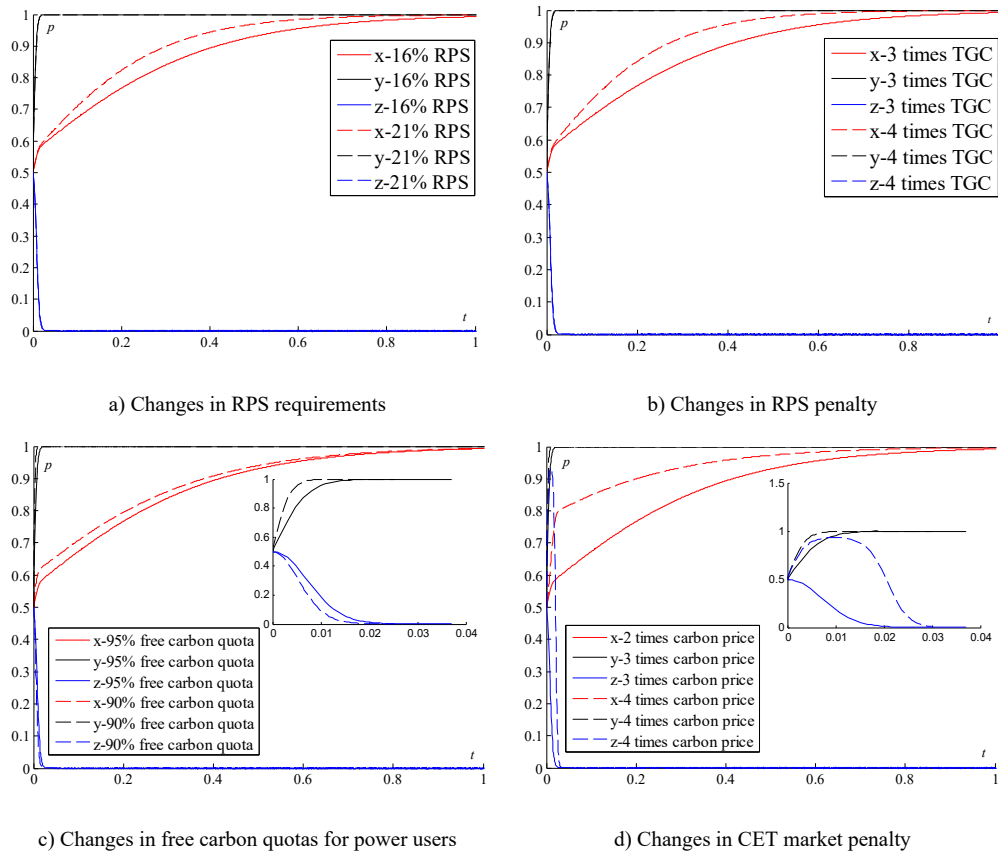


Fig 6-2 Sensitivity analysis of stable point E_7

It can be seen from Figure 6-2 a) and b) that by increasing the RPS requirement value or punishment, the evolution trend of the green power selection strategy of power users can be accelerated. It can be seen from Figure 6-2 c) and d) that by reducing the proportion of free carbon quota for power users and increasing the punishment of the CET market, it can also accelerate the evolution trend of power users' green power selection strategies. At the same time, it can accelerate the evolution trend of power users' participation in carbon market selection strategies. For thermal power manufacturers, reducing the proportion of free carbon quota for power users can accelerate the evolutionary trend of thermal power manufacturers not choosing to participate in the carbon market. Increasing the punishment of the CET market will make the CET market selection strategy of thermal power manufacturers evolve to participate in CET market first, and then gradually reverse and stabilize to not participate in CET market.

The above analysis shows that under the precondition of satisfying the system equilibrium state E_7 , when the policy parameters of RPS and CET are changed, except for the CET market penalty parameters, changes in other policy parameters can have a better directional guiding effect on the selection strategies of market participants.

6.3 Sensitivity analysis of E_2 , E_4 and E_6

This section continues the sensitivity analysis of three other evolutionary equilibrium states.

(1) When $F_c - P_h < 0$ and $P_t - P_g + \alpha F_{RPS} < 0$, $E_2(0,0,1)$, $E_4(0,1,0)$ are stable points.

Based on the basic parameters of Jilin Province in Table 5-2, changes were made to meet the current stable conditions, P_g was set to 0.49, and F_c was set to 50. The stability analysis from different initial points are shown in Figure 6-3:

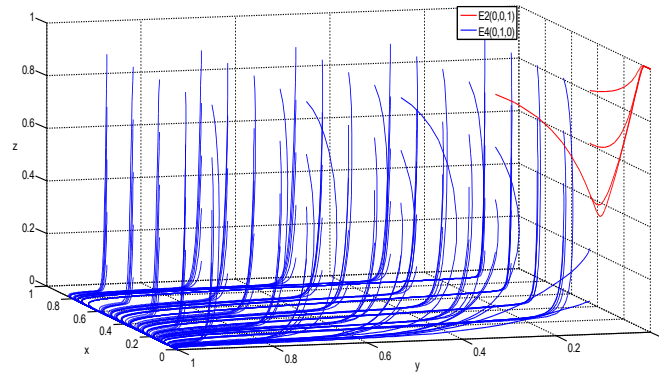


Fig 6-3 the influence of different initial points

It can be seen from Figure 6-3 that most of the initial point combinations evolve towards $E_4(0, 1, 0)$, and very few points evolve towards $E_2(0, 0, 1)$. From the numerical analysis results, When the initial ratio of x and y is too low (less than 0.3), the selection strategy of market players will eventually evolve to $E_2(0,0,1)$. Similarly, a sensitivity analysis is performed on the stable point $E_4(0, 1, 0)$, and the analysis results are shown in figure 6-4.

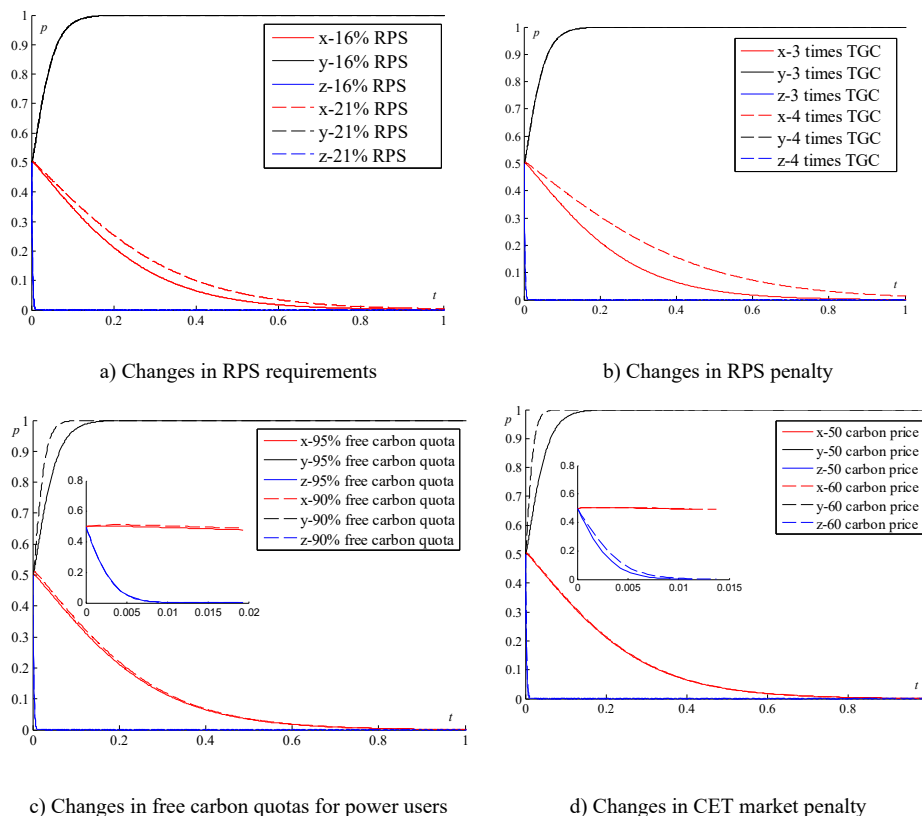
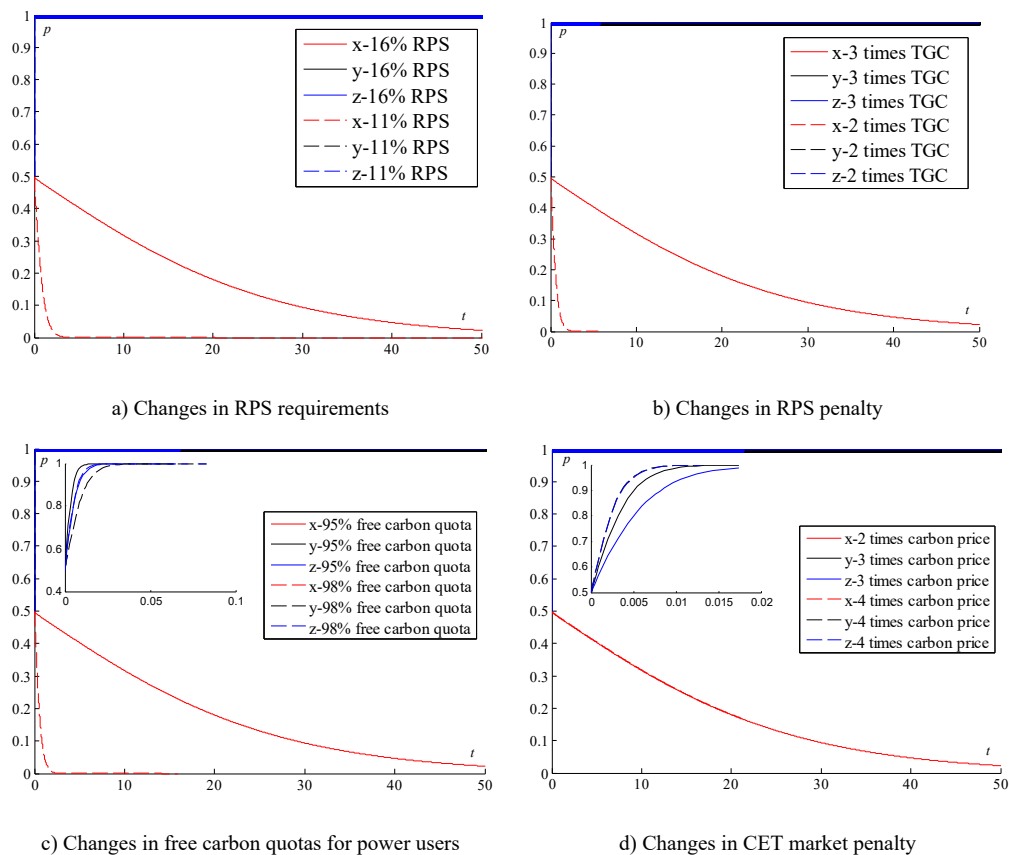


Fig 6-4 Sensitivity analysis of stable point $E_4(0,1,0)$

It can be seen from Figure 6-4 a) and b) that by increasing the RPS requirement value or punishment, it will slow down the evolution of power users to the strategy of not choosing green electricity. It can be seen from Figure 6-4 c) and d) that by reducing the the proportion of free carbon quota for power users and the increased punishment in CET market can accelerate the evolution trend of power users' participation in CET market selection.

The above analysis shows that under the precondition of satisfying the system equilibrium state E_4 , by changing the policy parameters of RPS and CET market, it can have a certain directional guiding effect for power

483 users' green power selection strategy and CET market selection strategy.
 484 (2) when $F_c - P_h > 0$ and $Q_e(P_t - P_g + \alpha F_{RPS}) + \theta E_e I_e Q_e(P_h - P_m) < 0$, $E_6(0,1,1)$ is a stable point. Changes are made based
 485 on the basic parameters of Jilin Province in Table 5-2, P_g is set to 0.49, F_c is set to 120, and P_h is set to 50. The
 486 sensitivity analysis is shown in figure 6-5.



487 Fig 6-5 Sensitivity analysis of stable point $E_6(0,1,1)$
 488 It can be seen from Figure 6-5 a) and b) that by increasing the RPS requirement or punishment, the evolution
 489 of power users to the strategy of not choosing green electricity will be slowed down; It can be seen from Figure 6-
 490 5 c) that increasing the proportion of free carbon allowances for power users will rapidly accelerate the evolution
 491 of power users to the strategy of not choosing green electricity, and will slow down the evolution of power users'
 492 carbon market selection strategy. It can be seen from Figure 6-5 d) that increasing the CET market penalty will
 493 accelerate the evolution of power users and thermal power manufacturers to participate in CET market.

494 The above analysis shows that, under the precondition of satisfying the system equilibrium state E_6 , modifying
 495 different policy parameters has different guiding effects on the strategic choices of various market participants.

496 6.4 System Equilibrium Stability Approach

497 According to the stable conditions of each equilibrium point and the numerical simulation results, if the policy
 498 parameter setting or the change range of the initial population proportion changes the combination of stable
 499 conditions in Table 4-2, the final equilibrium state of the system evolution will also change, its change process is
 500 shown in figure 6-6.

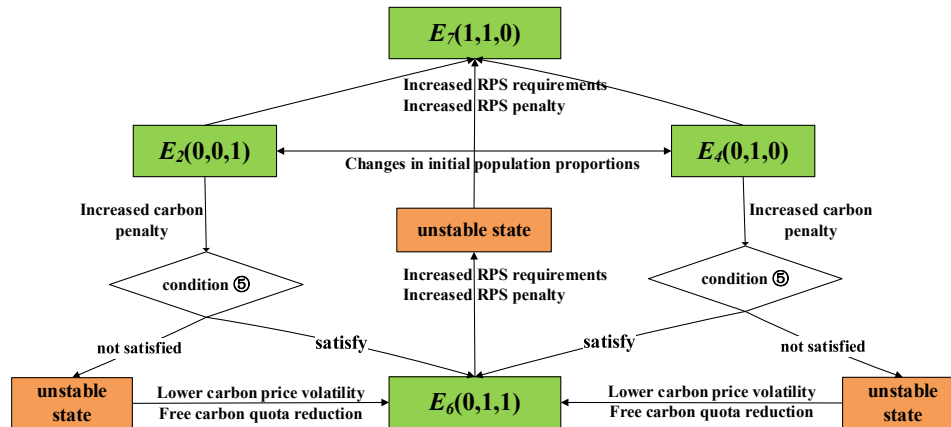


Fig 6-6 the process of changing the stable state

As shown in Figure 6-6, by changing the policy parameters, the stable state of the system can be changed between equilibrium points, but it is worth noting that if policy parameters changes do not completely change the combination of stable conditions, the system will fall into an unstable state.

Taking the $E_6(0,1,1)$ equilibrium point as an example, its stability conditions are $F_c - P_h > 0$ and $(P_t - P_g + \alpha F_{RPS}) + \theta E_e I_e (P_h - P_m) < 0$, when the RPS requirements increases or penalty increases, then the condition $(P_t - P_g + \alpha F_{RPS}) + \theta E_e I_e (P_h - P_m) > 0$ is satisfied, but the condition $P_t - P_g + \alpha F_{RPS} < 0$ is still satisfied. In this case, there is no matching combination of stable conditions, the system enters an unstable state, and only continues to increase the RPS requirements or penalty, so that the system can enter the stable state $E_7(1,1,0)$. The opposite is also true, taking the stable point $E_7(1,1,0)$ as an example, reducing the RPS requirements or penalty will cause the system to enter an unstable state first, and then enter the stable point $E_6(0,1,1)$.

In addition to changing the RPS and CET market-related parameters, changes in electricity prices also can influence the stable state. Taking $E_7(1,1,0)$ as an example, if the price of green power changes, it is assumed that the purchase of green power requires the purchase of the TGCs, this model is called the integration of electricity and certificates. At this time, the price of green power is $P'_g = P_g + F_{RPS}$. If the system is currently in the stable state $E_7(1,1,0)$, the adoption of the "integration of electricity and certificate" policy will make the system enter an unstable state, as shown in Figure 6-7:

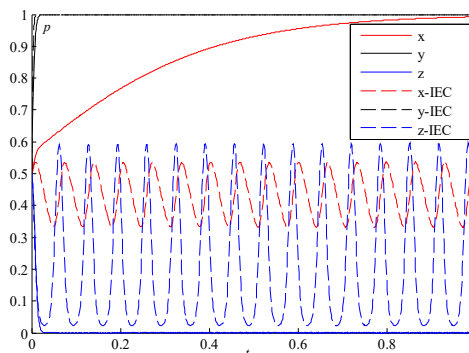


Fig 6-7 Evolution diagram after "integration of electricity and certificate" in the state of $E_7(1,1,0)$

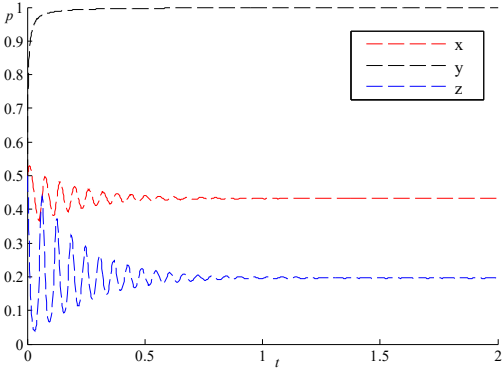
For the treatment of system instability, Fang et al. (2019)^[28] proposes a dynamic penalty mechanism to make the system gradually tend to a stable state. This paper adjusts the model according to this idea, and adopts a dynamic adjustment mechanism for the RPS requirement and free carbon quota, that is, as the proportion of power users

524 choosing green power strategies and the proportion of participating in the CET market increases, the RPS
 525 requirement and CET quota are gradually reduced. The formula is adjusted as follows:

526
$$\alpha = \alpha \left(\frac{1}{x} - 1 \right) \quad (4)$$

527
$$\theta = \theta \left(\frac{1}{y} - 1 \right) \quad (5)$$

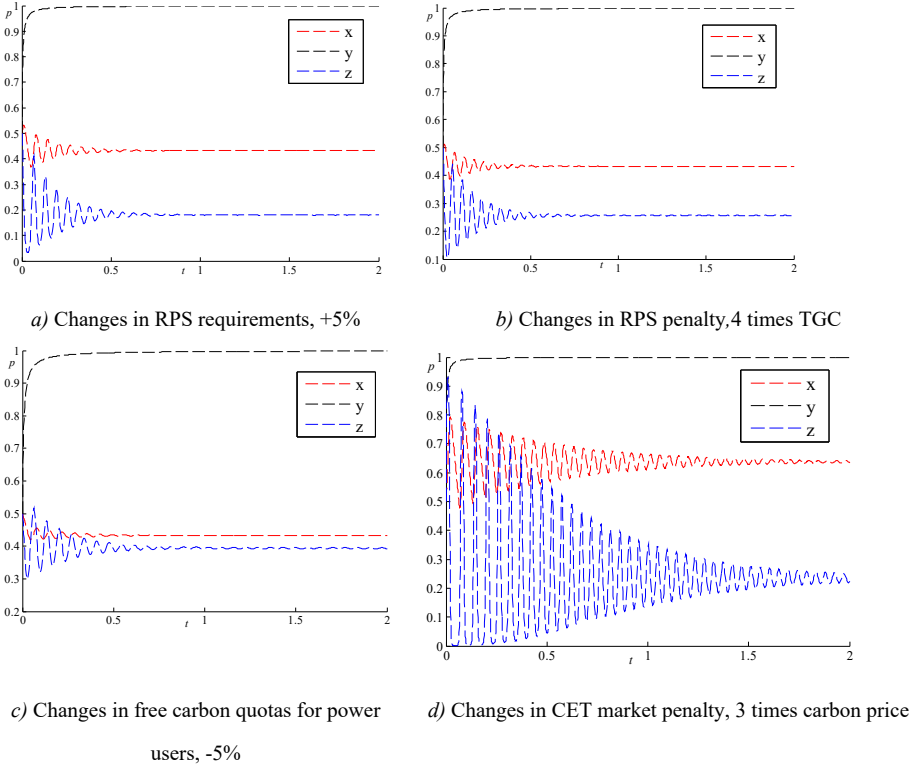
528 Based on equation (4) (5), update equation (1) (2) (3), the system evolution result after adopting the dynamic
 529 adjustment mechanism is shown in figure 6-8.



530
 531 Fig 6-8 Evolution under the dynamic adjustment mechanism

532 It can be seen from Figure 6-8 that the dynamic adjustment mechanism can make the system evolve to a stable
 533 state (0.4323, 1, 0.1952).

534 From the sensitivity analysis in Figures 6-9, it can be seen that adjusting the RPS-related parameters has a low
 535 sensitivity, and adjusting the carbon market-related parameters will have a greater impact on the green electricity
 536 purchase strategy of power users. Under the higher carbon quota penalty, the CET market selection strategy of
 537 thermal power manufacturers gradually converges from a relatively large volatility to a stable state.



From the overall analysis results, through the design of the dynamic adjustment mechanism, the unstable state of the system can be transformed to a stable state, but the final equilibrium point still fluctuates around the initial proportion of each group.

7 Conclusions and policy implications

This paper analyzes the impact of different policy parameters on the decision-making behavior of stakeholders when the RPS assessment is finally transmitted to the user side and the carbon market expands to industries other than the power generation industry.

Based on the operation mechanism of RPS and CET market, this paper constructs a multi-party evolutionary game model between power users and power generators, and analyzes the impact of different policy parameters on the decision-making behavior of stakeholders when the RPS assessment is finally transmitted to the user side and the CET market expands to industries other than the power generation industry. It provides an analytical framework for analyzing the guiding effect of different policy parameters on the strategies of market participants.

Through the analysis of China's actual operation data, this paper has the following findings:

(1) Under the synergistic effect of the dual market, the selection strategies of the market participants have an interaction relationship. The green electricity selection strategy and CET market selection strategy for power users have an opposite relationship, the CET market selection strategy of power users between thermal power manufacturers are a mutually exclusive relationship, and the green power selection strategy of power users and the CET market selection strategy of thermal power manufacturers have a mutually exclusive relationship.

(2) Under different combinations of policy parameters, the group strategies of market participants will evolve to different equilibrium and stable points, and are prone to fall into an unstable state during the evolution process.

(3) Through a fixed dynamic adjustment mechanism, the system can gradually evolve to a stable state, but the policy parameters have little impact on the stable state of the system and cannot show a good guiding effect, which the stable state only fluctuates near the initial state.

The following policy implications can be derived from the discussion in this paper:

(1) Although the RPS and CET markets belong to different government departments, when making policy formulations, it is necessary to consider the opposite and mutually exclusive influences between the two markets. We need to monitor the changes in the proportion of strategic choices of market players, track the evolution in the process of market development, and change reasonable policy parameters to promote the healthy development of the two markets.

(2) The policy parameters of the RPS and CET markets need to be set according to development needs. In order to solve the market risks caused by the uncertainty of market players' strategies, a dynamic adjustment mechanism can be adopted for adjustment; when focusing on the development of a certain type of market, the parameters can be set according to the combination of stability conditions mentioned above to be used for guide the selection strategies of market participants to develop in the expected direction.

(3) When government departments formulate policy parameters, such as RPS requirements, free carbon quota, assessment method, and punishment degree, they need to comprehensively consider the sensitivity of user policy choices to avoid the system from entering an unstable state, which can be qualitatively based on the analysis framework of this paper.

Author contribution

HU Yu: Writing-original draft, Data curation, Conceptualization, Methodology, Software. CHI Yuanyang:

Supervision, Writing-review & editing, Funding acquisition. ZHOU Wenbing: Writing-review & editing, Supervision. YUAN Yongke: Writing-review, Validation. ZHAO Hao: Writing-review, Data curation.

Declaration of competing interest

This study was not funded by any organization. All Authors declare that she has no conflict of interest. This article does not contain any studies with human participants performed by any of the authors.

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References

- [1] TU Qiang, MO Jianlei, FAN Ying. The evolution and evaluation of China's renewable energy policies and their implications for the future[J]. China population, resources and environment, 2020,30(3):29-36.(in Chinese)
- [2] NDRC. 2021. Notice on matters related to the renewable energy Feed-in tariff policy in 2021 [EB/OL]. https://www.ndrc.gov.cn/xxgk/zcfb/tz/202106/t20210611_1283088_ext.html.
- [3] BAO X, ZHAO W, WANG X, et al. Impact of policy mix concerning renewable portfolio standards and emissions trading on electricity market [J]. Renew. Energy, 2019, 135:761-774.
- [4] NDRC. 2019. Circular on establishing and improving the Guarantee mechanism for the consumption of renewable energy electricity [EB/OL]. https://www.ndrc.gov.cn/xxgk/zcfb/tz/201905/t20190515_962446.html.
- [5] MEE. 2021. Measures for the Administration of Carbon Emission Trading (for trial) [EB/OL]. http://www.gov.cn/zhengce/zhengceku/2021-01/06/content_5577360.htm.
- [6] Zhang H, Zhang B. The unintended impact of carbon trading of China's power sector [J]. Energy Policy. 2020, 147: 111876.
- [7] ZHU J, FAN Y, DENG X, et al. Low-carbon innovation induced by emissions trading in China [J]. Nat. Commun. 2019. <https://doi.org/10.1038/s41467-019-12213-6>.
- [8] Yu D, Li J. Evaluating the employment effect of China's carbon emission trading policy: Based on the perspective of spatial spillover [J]. J. Cleaner Prod. 2021, 292: 126052.
- [9] ZHANG H, DUAN M, DENG Z. Have China's pilot emissions trading schemes promoted carbon emission reductions? the evidence from industrial sub-sectors at the provincial level [J]. J. Cleaner Prod.2019, 234: 912 — 924.
- [10] YAN Y, ZHANG X, ZHANG J, et al. Emissions trading system (ETS) implementation and its collaborative governance effects on air pollution: the China story [J]. Energy policy, 2020, 138:111282.
- [11] Qi S, Zhou C, Li K, et al. The impact of a carbon trading pilot policy on the low-carbon international competitiveness of industry in China: An empirical analysis based on a DDD model [J]. J. Cleaner Prod. 2021, 281.
- [12] Wang Wenju, Chen Zhenling. Research on the Initial Carbon Allowance Allocation Scheme in China's Provincial Regions: Based on the Perspectives of Responsibility and Objectives, Equity and Efficiency [J]. Management World. 2019, 35(03):81-98. <https://doi.org/10.19744/j.cnki.11-1235/f.2019.0036>. (in Chinese)
- [13] Barbose G, Wiser R, Heeter J, et al. A retrospective analysis of benefits and impacts of U.S. renewable portfolio standards [J]. Energy Policy. 2016, 96: 645-660.
- [14] Liu X, Jin Z. An analysis of the interactions between electricity, fossil fuel and carbon market prices in Guangdong, China [J].

Energy Sustainable Dev. 2020, 55: 82-94.

- [15] Schusser S, Jaraité J. Explaining the interplay of three markets: Green certificates, carbon emissions and electricity [J]. *Energy Econ.* 2018, 71: 1-13.
- [16] Xin-Gang Z, Yu-Zhuo Z, Ling-Zhi R, et al. The policy effects of feed-in tariff and renewable portfolio standard: A case study of China's waste incineration power industry [J]. *Waste Manage.* 2017, 68: 711-723.
- [17] Zhang Y, Zhao X, Ren L, et al. The development of the renewable energy power industry under feed-in tariff and renewable portfolio standard: A case study of China's wind power industry [J]. *J. Cleaner Prod.* 2017, 168: 1262-1276.
- [18] Feng, Tian-tian, Yang, Yi-sheng, Yang, YH. What will happen to the power supply structure and CO2 emissions reduction when TGC meets CET in the electricity market in China? [J]. *Renewable Sustainable Energy Rev.* 2018, 92:121-132.
- [19] Zhou Ying, Zhao Xin-gang. The impact of Renewable Portfolio Standards on carbon emission trading under the background of China's electricity marketization reform[J]. *Energy*, 2021:226.
- [20] Weng Q, Xu H. A review of China's carbon trading market [J]. *Renewable Sustainable Energy Rev.* 2018, 91: 613-619.
- [21] Dai H, Xie Y, Liu J, et al. Aligning renewable energy targets with carbon emissions trading to achieve China's INDCs: A general equilibrium assessment [J]. *Renewable Sustainable Energy Rev.* 2018, 82: 4121-4131.
- [22] Chen W, Chen J, Ma Y. Renewable energy investment and carbon emissions under cap-and-trade mechanisms [J]. *J. Cleaner Prod.* 2021, 278: 123341.
- [23] Currier K M. A regulatory adjustment process for the determination of the optimal percentage requirement in an electricity market with Tradable Green Certificates [J]. *Energy Policy.* 2013, 62: 1053-1057.
- [24] Tanaka M, Chen Y. Market power in renewable portfolio standards [J]. *Energy Econ.* 2013, 39: 187-196.
- [25] Sun Y. The optimal percentage requirement and welfare comparisons in a two-country electricity market with a common tradable green certificate system [J]. *Econ. Model.* 2016, 55: 322-327.
- [26] Helgesen P I, Tomasgard A. An equilibrium market power model for power markets and tradable green certificates, including Kirchhoff's Laws and Nash-Cournot competition [J]. *Energy Econ.* 2018, 70: 270-288.
- [27] Zhao X, Ren L, Zhang Y, et al. Evolutionary game analysis on the behavior strategies of power producers in renewable portfolio standard [J]. *Energy.* 2018, 162: 505-516..
- [28] Fang D, Zhao C, Kleit A N. The impact of the under enforcement of RPS in China: An evolutionary approach [J]. *Energy Policy.* 2019, 135: 111021.
- [29] Bonenti F, Oggioni G, Allevi E, et al. Evaluating the EU ETS impacts on profits, investments and prices of the Italian electricity market [J]. *Energy Policy.* 2013, 59: 242-256.
- [30] Yi B, Xu J, Fan Y. Coordination of policy goals between renewable portfolio standards and carbon caps: A quantitative assessment in China [J]. *Appl. Energy.* 2019, 237: 25-35.
- [31] Xu J, Lv T, Hou X, et al. Provincial allocation of renewable portfolio standard in China based on efficiency and fairness principles [J]. *Renew. Energy.* 2021, 179: 1233-1245.
- [32] Pineda S, Bock A. Renewable-based generation expansion under a green certificate market [J]. *Renew. Energy.* 2016, 91: 53-63.
- [33] Suo C, Li Y P, Jin S W, et al. Identifying optimal clean-production pattern for energy systems under uncertainty through introducing carbon emission trading and green certificate schemes [J]. *J. Cleaner Prod.* 2017, 161: 299-316.
- [34] Guo H, Chen Q, Xia Q, et al. Modeling Strategic Behaviors of Renewable Energy With Joint Consideration on Energy and Tradable Green Certificate Markets[J]. *IEEE Trans. Power Syst.* 2020, 35(3): 1898-1910.
- [35] Weng Q, Xu H. A review of China's carbon trading market [J]. *Renewable Sustainable Energy Rev.* 2018, 91: 613-619..
- [36] NEA. 2020 National Renewable Energy power development Monitoring and Evaluation Report [R]. Beijing, 2020.
- [37] ETC. China 2050: a fully developed rich zero-carbon economy[R]. Beijing, 2019.
- [38] LI Zheng, CHEN Siyuan, DONG Wenjuan, et al. Low Carbon Transition Pathway of Power Sector under Carbon

-
- Emission Constraints [J]. Proceedings of the CSEE.2021; 1-15. <https://doi.org/10.13334/j.0258-8013.pcsee.210671>.(in Chinese)
- [39] Xie Shiyu. Economic Game Theory (4th Edition)[M]. Shanghai. 2019:261-262.
- [40] .NDRC, NEA. 2021. Notice on the weight of responsibility for the consumption of renewable energy power in 2021 and related matters [EB/OL].http://www.gov.cn/zhengce/zhengceku/2021-05/26/content_5612441.htm.
- [41] NDRC. 2020. Notice on matters related to the photovoltaic feed-in tariff policy in 2020 [EB/OL].2020-03-31[2021-10-13]. <http://www.jianghai.gov.cn/attachment/0/150/150439/2173982.pdf>.
- [42] NDRC. 2019. Notice on improving the policy of wind power on-grid pricing [EB/OL]. http://www.sgzy.gov.cn/sgzyfgj/gkmlpt/content/0/702/post_702185.html#3590.
- [43] SHI Xueying, YANG Bowen. Penalty rule design for emissions control firms not achieving carbon trading performance goals [J]. China population, resources and environment, 2018, 28(4) :35—42.(in Chinese)
- [44] MEE. 2020. Implementation Plan on Setting and Allocation of Total Carbon Emission Trading Quota in China in 2019-2020 (Power Generation Industry) [EB/OL].https://www.mee.gov.cn/xxgk/xxgk03/202012/t20201230_815546.html.
- [45] China Electricity Council. 2021. China Electric Power Industry Annual Development Report 2021 [R].Beijing.
- [46] Shanghai Municipal Bureau of Ecology and Environment. 2020. Notice on the issuance of the List of Entities included in The Management of Carbon Emission Quota of Shanghai Municipality (2020 Edition) and the Allocation Plan of Carbon Emission Quota of Shanghai Municipality in 2020[EB/OL]. <https://sthj.sh.gov.cn/hbzhwypt2025/20210202/510b31e87df149348d73c7a40faab484.html>.
- [47] TONG Qing, JIANG Dongmei, WEI Xinyang. Greenhouse Gas Accounting Method and Case Analysis of Electrolytic Aluminum Enterprises Covered by China Carbon Market [J]. Ecological economy. 2019, 35(03):23-26. (in Chinese)

Appendix A

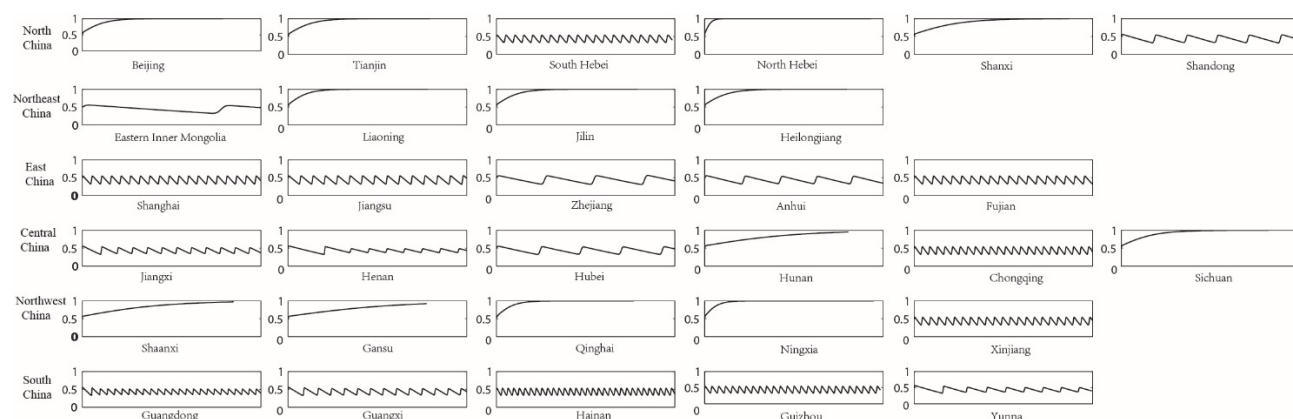


Figure A.1 electricity selection evolution for power users

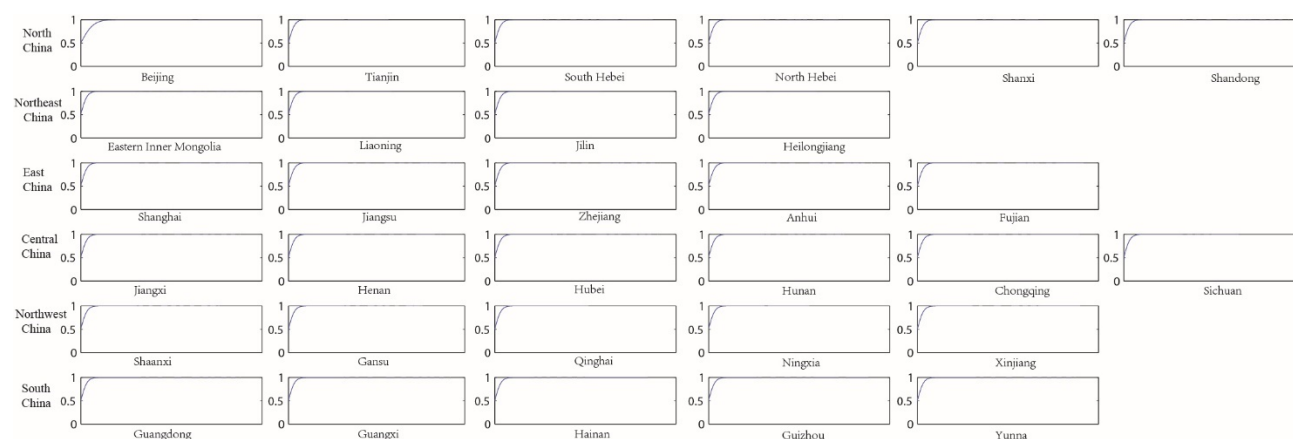


Figure A.2 Evolution of carbon trading strategies for power users

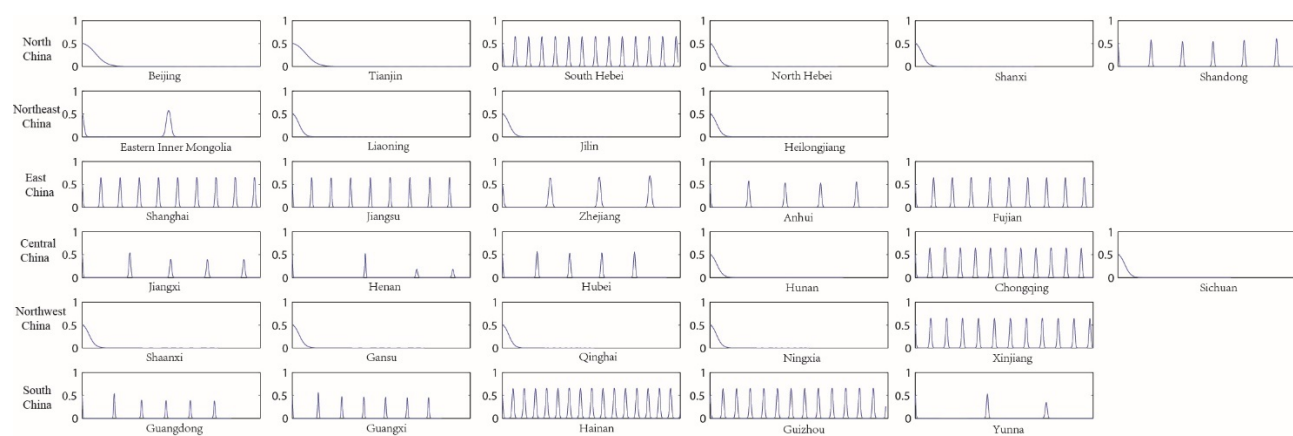


Figure A.3 Evolution of carbon trading strategies for thermal power manufacturers

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Appendix B

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Table B.1 Numerical calculation results of the equilibrium points of E_{10} - E_{14}

equilibrium solution	southern Hebei			Jilin		
	value	eigenvalues	stability	value	eigenvalues	stability
E_{10}	(0, 3.1986, 3.1986)	/	Solution over bounds	(0, 3.1986, 3.1986)	/	Solution over bounds
E_{11}	(0.4322, 1, 0.1374)	(-64.4772i, 64.4772i, -321.53 + 1.0658e-14i)	Solution over bounds	(0.4322, 1, -0.0654)	/	Solution over bounds
E_{12}	(-3.1886, 0.1374, 1)	/	Solution over bounds	(-3.1886, 0.1374, -0.0654)	/	Solution over bounds
E_{13}	(-280.4494, 8.3506, 0.0165)	/	Solution over bounds	(596.4613, 8.3899, -0.0078)	/	Solution over bounds
E_{14}	(0.5509, 0.0266, 5.1578)	/	Solution over bounds	(0.5551, -0.0126, 5.1820)	/	Solution over bounds

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Appendix C

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Table C.1 Basic data

province	Minimum RPS	RPS incentive value	On-grid tariff of thermal power (Yuan/kWh)	Wind power division	On-grid tariff of wind power (Yuan/kWh)	PV zoning	光伏上网电价 (Yuan/kWh)
Beijing	17.50%	19.30%	0.3598	I、III	0.335	II	0.4
Tianjin	16.00%	17.60%	0.3655	IV	0.47	II	0.4
south Hebei	16.00%	17.60%	0.3644	II	0.34	III	0.49
north Hebei	16.00%	17.60%	0.372	IV	0.47	II	0.4
Shanxi	19.00%	20.90%	0.332	IV	0.47	II、III	0.445
Shandong	12.50%	13.80%	0.3949	IV	0.47	III	0.49
western inner Mongolia	19.50%	21.50%	0.3035	I	0.29	II	0.4
eastern inner Mongolia	19.50%	21.50%	0.2829	IV	0.47	I	0.35
Liaoning	13.50%	14.90%	0.3749	IV	0.47	II	0.4
Jilin	21.00%	23.10%	0.3731	III、IV	0.425	II	0.4
Heilongjiang	20.00%	22.00%	0.374	III、IV	0.425	II	0.4
Shanghai	4.00%	4.40%	0.4155	IV	0.47	III	0.49
Jiangsu	10.50%	11.60%	0.391	IV	0.47	III	0.49
Zhejiang	8.50%	9.40%	0.4153	IV	0.47	III	0.49
Anhui	14.00%	15.40%	0.3844	IV	0.47	III	0.49
Fujian	7.50%	8.30%	0.3932	IV	0.47	III	0.49
Jiangxi	12.00%	13.20%	0.4143	IV	0.47	III	0.49
Henan	18.00%	19.80%	0.3779	IV	0.47	III	0.49
Hubei	10.00%	11.00%	0.4161	IV	0.47	III	0.49

Hunan	13.50%	14.90%	0.45	IV	0.47	III	0.49
Chongqing	4.00%	4.40%	0.3964	IV	0.47	III	0.49
Sichuan	6.00%	6.60%	0.4012	IV	0.47	II	0.4
Shaanxi	15.00%	16.50%	0.3545	IV	0.47	II 、 III	0.445
Gansu	18.00%	19.80%	0.3078	II 、 III	0.36	I 、 II	0.375
Qinghai	24.50%	27.00%	0.3247	IV	0.47	I 、 II	0.375
Ningxia	22.00%	24.20%	0.2595	III	0.38	I	0.35
Xinjiang	12.50%	13.80%	0.25	IV	0.47	I 、 II	0.375
Guangdong	5.00%	5.50%	0.453	IV	0.47	III	0.49
Guangxi	10.00%	11.00%	0.4207	IV	0.47	III	0.49
Hainan	8.00%	8.80%	0.3515	IV	0.47	III	0.49
Guizhou	8.50%	9.40%	0.3515	IV	0.47	III	0.49
Yunnan	15.00%	16.50%	0.3358	II	0.34	II	0.4

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